HDOC Day-48 Activity

C Program to Find the Pivot Index of an Array Using Pointers

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Aim

To write a C program that reads an integer array and finds the leftmost pivot index using a function and pointers. A pivot index is an index where the sum of elements strictly to its left equals the sum of elements strictly to its right.

Algorithm

1. Read the size of the array.

2. Read all array elements.

3. Pass the array address and size to the pivot_index() function.

4. Calculate the total sum of all elements.

5. Traverse the array and calculate rightSum = total - leftSum - current element.

6. If leftSum equals rightSum, return the current index.

7. Otherwise, add the current element to leftSum and continue.

8. If no pivot index is found, return -1.

C Program

```
#include <stdio.h>

int pivot_index(int *arr, int size)
{
    int total = 0;
    int leftSum = 0;

    for (int i = 0; i < size; i++)
    {
        total += *(arr + i);
    }

    for (int i = 0; i < size; i++)
    {
        int rightSum = total - leftSum - *(arr + i);

        if (leftSum == rightSum)
        {
            return i;
        }

        leftSum += *(arr + i);
    }

    return -1;
}

int main()
{
    int size;

    printf("Enter the size of array: ");
    scanf("%d", &size);
```

```
    if (size <= 0)
    {
        printf("Invalid array size\\n");
        return 0;
    }

    int arr[size];

    printf("Enter the array elements: ");
    for (int i = 0; i < size; i++)
    {
        scanf("%d", &arr[i]);
    }

    int result = pivot_index(arr, size);

    printf("Pivot index: %d\\n", result);

    return 0;
}
```

Sample Input and Output

Input	Output
6 1 7 3 6 5 6	Pivot index: 3

Explanation

For the array 1, 7, 3, 6, 5, 6, at index 3 the left sum is 1 + 7 + 3 = 11 and the right sum is 5 + 6 = 11. Since both sums are equal, the pivot index is 3. The first element has a left sum of 0, and the last element has a right sum of 0.

Viva Questions

1. What is a pivot index? An index where the sum of elements on the left equals the sum of elements on the right.

2. What is a pointer? A variable that stores the memory address of another variable.

3. Why is int *arr used? It lets the function receive the address of the first array element.

4. What does *(arr + i) mean? It accesses the element at index i using pointer arithmetic.

5. Why return -1? It indicates that no pivot index exists.

6. What is the time complexity? O(n), because the array is traversed twice.